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Project Title:	Dose-rate effects of protons on the induction of genomic instability in vivo		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
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Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Space Radiation NNJ07ZSA001N
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
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	Protons are the most abundant component of solar particle events (SPEs) in space. Little is known about early or late occurring biological effects of in vivo exposure to protons at doses and low dose rates that are similar to what astronauts encounter in space. During a journey to Mars, every cell in the body will be hit by a proton once every three days, and by a heavy ion approximately once a month. One of the major concerns associated with the exposure of the astronauts to protons during space flights is cancer induction. To protect astronauts in space, we must understand the radiation-induced cellular and molecular changes that influence cancer induction. Since in vivo studies using humans are not possible, animal studies are critically important. This study is designed to determine the dose-rate effects of protons (100 MeV/amu) on in vivo induction of genomic instability (expressed as delayed chromosomal damage). This approach is highly relevant for assessing cancer risks;		

Task Description:	however, it has not been applied with proton exposure. We will characterize the type, the frequency, and the distribution of initial and subsequent chromosome aberrations (CAs), and the expression of genomic instability induced in vivo in bone marrow (BM) cells after exposure of mice to 1GeV/amu protons at the dose and dose rates encountered in space. A novel genome-wide multicolor fluorescence in situ hybridization (mFISH) of mouse chromosomes will be used to detect and characterize chromosome aberrations in metaphase cells prepared from BM cells of exposed mice collected at different times (up to 6 months) post-irradiation. To increase the reliability of risk assessment, a much improved understanding of the molecular basis for dose-rate effects of protons is needed. Therefore, this research project is also designed to determine the potential molecular basis for the dose-rate dependencies of proton-induced genomic instability in vivo. To achieve this goal, we will carefully measure levels of activated nuclear factor-kappaB (NF-kappaB), a highly important transcription factor (a protein that controls the transcription of the genes), and the expression of selected NF-kappaB target proteins involved in cell survival and inflammatory response in BM cells collected at different times after irradiation. Our research will be the first to use integrative in vivo endpoints in the tissue (BM) that is also the target of radiation-induced myeloid leukemia to evaluate the dose-rate effects of protons. Our data will not only contribute to reducing uncertainties in the assessment of cancer risk from exposure to radiation during space flights but will also improve the understanding of the molecular basis for Linear Energy Transfer (LET) dependencies of genomic instability. The results will provide high-priority information for NASA, because it has been widely suggested that elevation of genomic instability also elevates accumulation of genetic alterations that, in turn, increase cancer risk.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	New exploration of Moon and Mars by the US-NASA will undoubtedly happen in the very near future. It is therefore important to learn more about potential health effects of space radiation before the manned missions to these other worlds become a routine adventure. Such information should help reduce the uncertainties associated with risk assessment for human space exploration. Upon completion of the in vivo work proposed in this study, our data will definitely provide pivotal information that should lead to a better understanding of dose-rate effects and of mechanisms by which proton-induced aberrations can be potentially involved in the induction of genomic instability and cancer. The results also would facilitate determining whether there are unique biological effects associated with in vivo exposure to protons. Further, this study is important for understanding risks associated with space travel as well as developing advanced cancer therapies for Earth because it will couple molecular mechanisms with quantitative data from a robust, highly sensitive technique.
Task Progress:	We participated in several NSRL-BNL runs for proton-irradiation, i.e. NSRL-08BC, NSRL-09A, and NSRL-09BB. It has been confirmed that the beam time for proton-irradiation will be available for us in the fall run (NSRL-09C). For the reference radiation (137Cs gamma rays, 1 Gy delivered at 1 cGy/min), we have completed the irradiation (1 Gy, delivered at 1 cGy/min) using a gamma source at the Controlled Environment Radiation Facility (CERF) of Biology Department, BNL. It is necessary to include 137Cs gamma rays as a reference radiation because information on in vivo biological effects of 100 MeV/amu protons is scant. We used the mFISH method to determine the frequencies of chromatid- (G2-) type aberrations induced in BM cells collected at 3 hr after exposure of mice to either 1 Gy of 100 MeV/n protons or 1 Gy of 137Cs gamma rays (delivered at the same dose rate of 1 cGy/min). We found that several types of G2-aberrations in BM cells collected from mice exposed to either type of radiation. These were chromosomal exchanges and breaks (both chromatid- and chromosome-types). Chromatid exchanges included sister union (SU), and translocations [reciprocal translocations, Robertsonian (RT) or centric fusion, and incomplete types]. The results from our project demonstrate that in vivo exposure of mice to a single dose of 100 MeV/amu protons (delivered at a dose rate of 1 cGy/min) causes chromosomal damage in BM cells (the target cell for radiation-induced myeloid leukemia) collected at early time (3 hr post-irradiation). The frequency and the type of chromosomal damage are similar to those induced by a single dose of 1 Gy of 137Cs gamma rays, delivered at the same dose rate. It will be important to determine whether chromosomal damage will persist at late time following exposure to 100 MeV/amu protons. The results from such studies will provide high priority information for NASA, because it has been widely suggested that elevation of late-occurring chromosome instability (genomic instability) also elevates an accumulation of genetic alterations that, in turn, increases cancer risk. Although the mFISH method is robust and highly sensitive, it is very time consuming and laborious. Due to a very large number of metaphase cells to be scored, the evaluation of genome-wide CAs by mFISH will continue to be a major effort for years 3 and 4 of the study. We work diligently to generate data on the potential induction of genomic instability (measured by late-occurring CAs) by protons and gamma rays for NASA. Our study is important because it has been widely suggested that elevation of genomic instability also elevates accumulation of genetic alterations that, in turn, increase cancer risk. We have shown that a single dose of 1 Gy of 100 MeV/amu (delivered at 1 cGy/min) is capable of inducing NF-kappa B activation in BM cells of exposed mice. A high level of activated NF-kappa B was observed at 1.5 hr post-irradiation. Subsequently, the level of activated NF-kappa B returned to a normal level. Further, a significantly high level of TNF-alpha was detected at 1.5 and 24 hr (not at 3 hr) post-irradiation, while a significantly elevated level of IL-6 was observed at 3 hr post-irradiation only. In contrast, our results demonstrated that radiation exposure suppressed the expression level of IL-10 (an anti-inflammatory cytokine). These results provided a measure of the sensitivity of BM cells after in vivo exposure to 100 MeV/amu protons. Further, our findings indicated that the kinetics of expression of various cytokines after irradiation were different. Whether or not abnormal activation or expression of NF-kappa B and targeted proteins will be detected at later times (up to 6 months) post-irradiation remain unclear. Determinations of these endpoints in BM cells collected at later times and in BM cells collected from mice exposed to the same dose of 100 MeV/amu protons delivered at a lower dose rate (i.e. 0.5 cGy/min) are ongoing.
Bibliography Type:	Description: (Last Updated: 01/20/2026)

Abstracts for Journals and Proceedings	Rithidech KN, Honikel L, Reungpatthanaphong P, Simon SR, Whorton EB. "Early cellular and molecular responses after in vivo exposure to 100 MeV/amu protons." To be presented at the Heavy Ions in Therapy and Space Symposium, Cologne, Germany, July 6-10, 2009. Heavy Ions in Therapy and Space Symposium, Cologne, Germany, July 6-10, 2009. , Jul-2009
Abstracts for Journals and Proceedings	Rithidech KN, Honikel L, Simon SR, Spitz D, Whorton EB, Anderson CA. "Dose-rate effects of protons on the induction of genomic instability in vivo." Presented at the NASA Human Research Program Investigators' Workshop, League City, TX, February 2-4, 2009. NASA Human Research Program Investigators' Workshop, League City, TX, February 2-4, 2009. , Feb-2009
Articles in Peer-reviewed Journals	Rithidech KN, Supanpaiboon W, Honikel L, Whorton EB. "Induction of genomic instability after an acute whole-body exposure of mice to 56Fe ions." Adv Space Res, In Press, Corrected Proof, Available online 9 May 2009. http://dx.doi.org/10.1016/j.asr.2009.05.001 , May-2009
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